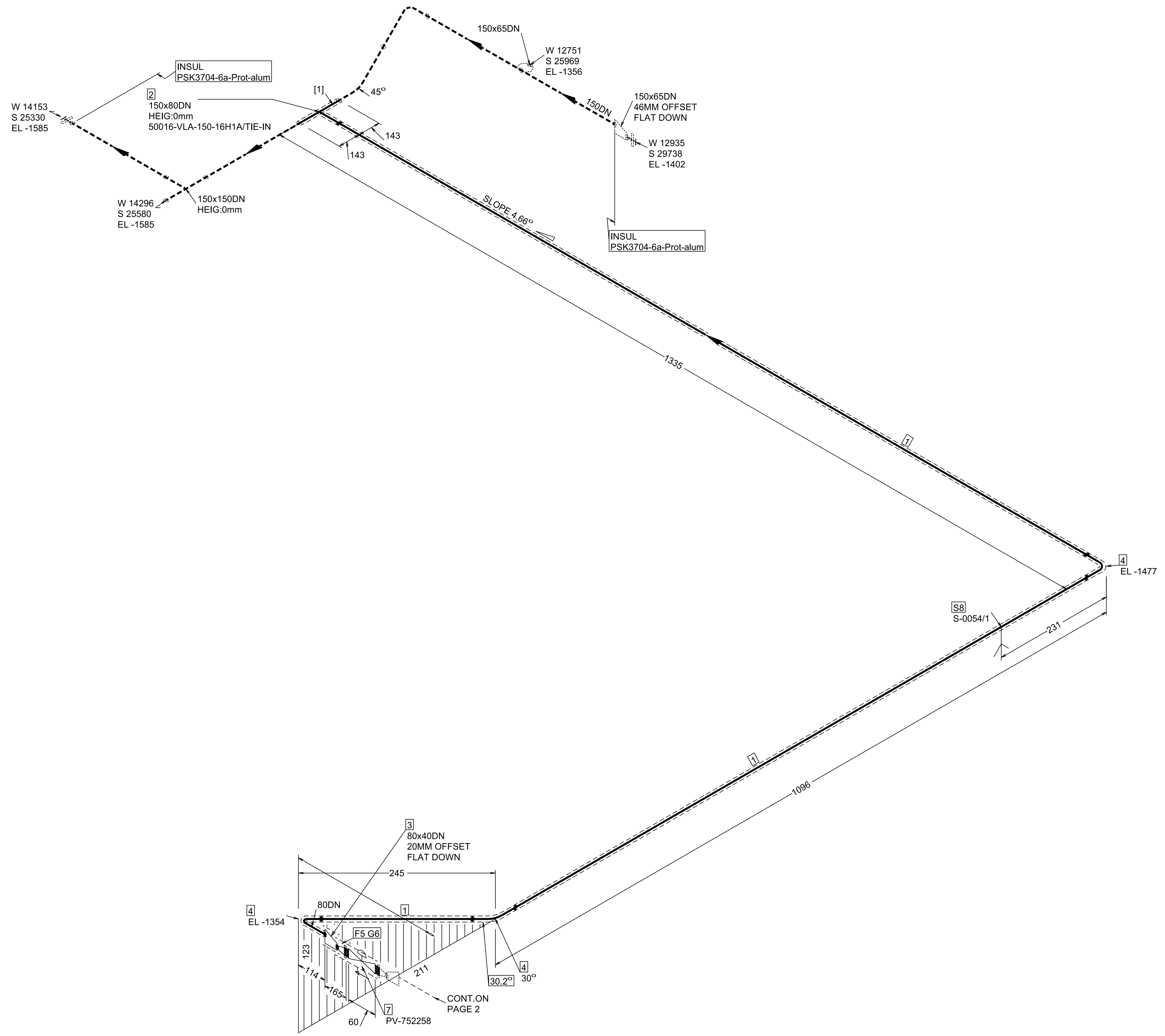
Project: VKB Zone: PD1-BM/A/DESI/PIPE/VLA 21/05/2026 DB: PREPAA/PREPAA Session: 4455 MDB: /119440.005



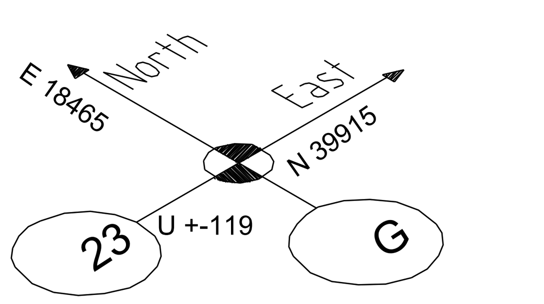
NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
1	Tube - 88.9 x 2.00 - EN 10217-7 - 1.4307-TC1	80	119440-VE16H1A-R00	2148MM
2	Reduced tee - w - EN 10253-4 - type A - Table A.2 - 168.3 x 3.00 / 88.9 x 3.00 - 1.4307 50016-VLA-150-16H1A/TIE-IN	150 x 80	119440-VE16H1A-R00	1
3	Eccentric reducer - w - EN 10253-4 - type A - Table A.6 - 88.9 x 2.00 / 48.3 x 2.00 - 1.4307	80 x 40	119440-VE16H1A-R00	1
4	Elbow - w - EN 10253-4 - type A - Table A.1 - model 3D - 90deg - 88.9 x 2.00 - 1.4307	80	119440-VE16H1A-R00	3
5	Weld-neck flange - EN 1092-1 - type 11 - B1 - 48.3/DN40 - PN40 - 1.4307	40	119440-VE16H1A-R00	1

NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
6	Gasket - EN 1514-1 - type IBC - OD48.3/DN40 - 1.5mm Thk - PN40 Fiber	40	119440-VE16H1A-R00	1
7	KPK BASIC BALL SECTOR VALVE WAFER PN40 SINGLE ACTING ACTUATOR PV-752258	40	KBE/INST01	1
	SUPPORTS			
8	HANGER PSK 7303 TYPE 12 See supportlist	80	VOFSUPP-PRIM-PSK-ATTA	1

PIPE SPOOLS

[1]

ORIGO & MARKINGS



1 DENOTES PARTS LIST
1 DENOTES WELD NO.
1 DENOTES REVISION
INSULATION
INSUL. + TRACE.
BRANCH START POINT

Insulation table (lengths according SSG 7645 Alt2)			
Bore	Thickness	Length	Insulation spec name
50	20mm	5897mm	/PSK3704-6a-Prot-alum
40	20mm	153mm	/PSK3704-6a-Prot-alum
80	20mm	2911mm	/PSK3704-6a-Prot-alum
15	20mm	182mm	/PSK3704-6a-Prot-alum
150	60mm	286mm	/PSK3704-6a-Prot-alum

NOTE:

Site weldings to be added by contractor.

Buildings construction and equipment erection tolerances to be checked by contractor.

Pipe levels and dimensions are according to pipe center line.

Isometric drawings for sizes DN 15-25 are approximate.
Piping contractor is responsible for final routing and supporting.

Welding factor	Acc. to pipe class
Effiency of long. weld	Acc. to pipe class
NDT circ. weld %	-
NDT long. weld %	-
Welding list see doc.	-

		Hazardous CLP content	
Directive	2014/68/EU	Pipe Class	/119440-VE16H1A-R00
Design press. (min)	-1bar	PED Category	SEP
Design temp. (min)	-	Media	
Max. allow. press. (PS)	-	Standard	
Operating press.	0.1bar	P&ID	RAU8F00303
Operating temp.	103degC	Design press.	6bar
PED Module	-	Design temp.	220degC
PED Group	-	Test press.	-
Wind loads	-	Pipe list doc. ID	RAUAF04300
External loads	-	Valve list doc. ID	RAUAF04301

Connection to existing pipe to be checked on site
For installation tolerances dimensions to be checked by pipe contractor at site.
Degree of prefabrication to be decided by contractor.

Rev	Revision Note	Created by	Checked by	Approved by	Date
0		J. Lakka	T. Saarinen	M. Kankkunen	5/5/2026

Site: Tervasaari, Valkeakoski	Equipment Type: Höyry- ja lauhdejärjestelmä		Equipment ID
Project Key: TERPM5_N5265A	Project Name: Ter PM 5 Safe Future	Project Number: N5265A	Work
Projection Method	Welding Symbols: ISO 2553	General Tolerances: ISO 13 920-BF ISO 2768-mK-E ISO 2768-mK-E ISO 2768-mK-E ISO 2768-mK-E	Scale: - / Size: A1 / Weight: -

Valmet

Title
50216-VLA-80-16H1A

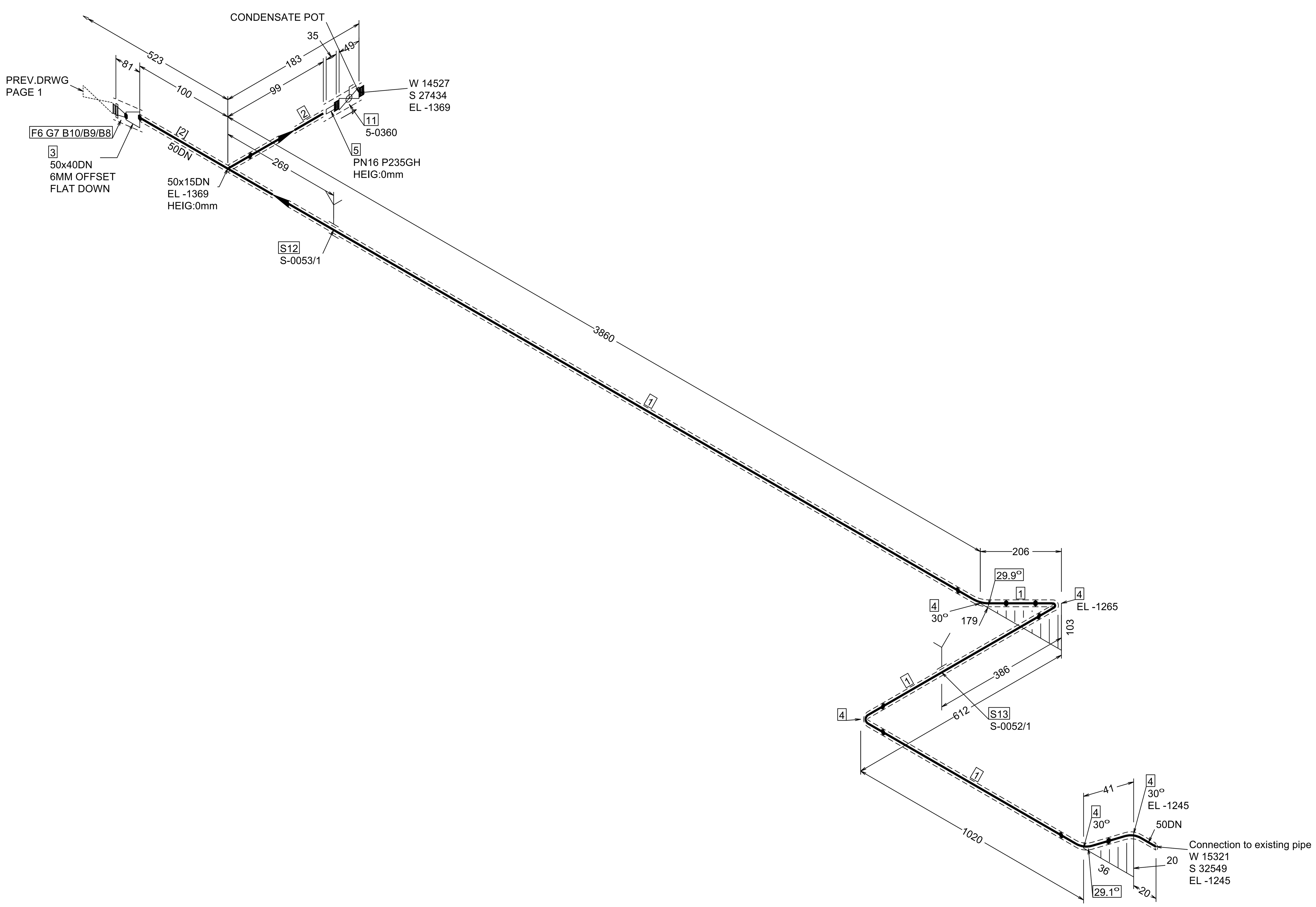
Status: CERTIFIED	Document ID	Sheet: 1/ 2
Security Classification: Internal		Revision: 0

NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
1	Tube - 60.3 x 2.00 - EN 10217-7 - 1.4307-TC1	50	119440-VE16H1A-R00	5434MM
2	Tube - 21.3 x 2.00 - EN 10217-7 - 1.4307-TC1	15	119440-VE16H1A-R00	70MM
3	Eccentric reducer - w - EN 10253-4 - type A - Table A.6 - 60.3 x 2.00 / 48.3 x 2.00 - 1.4307	50 x 40	119440-VE16H1A-R00	1
4	Elbow - w - EN 10253-4 - type A - Table A.1 - model 3D - 90deg - 60.3 x 2.00 - 1.4307	50	119440-VE16H1A-R00	5
5	STRAIGHT NIPPLE OTOE/PLAIN OTHER END NPT THRD. R-208 SHORT 316 PN16 P235GH	15	KBE/BBB001	1
6	Weld-neck flange - EN 1092-1 - type 11 - B1 - 48.3/DN40 - PN40 - 1.4307	40	119440-VE16H1A-R00	1

NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
7	Gasket - EN 1514-1 - type IBC - OD48.3/DN40 - 1.5mm Thk - PN40 Fiber	40	119440-VE16H1A-R00	1
8	130 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	16		4
9	Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG	16		4
10	Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG	16		8
11	KPK BASIC BALL VALVE TF/TF WI HAND LEVER 5-0360	15	KBE/VALV01	1
	SUPPORTS			
12	GUIDED SLIDE PSK 7303 TYPE 2, LOW See supportlist	50	VOFSUPP-PRIM-PSK-ATTA	1
13	SLIDE SUPPORT PSK 7303 TYPE 1, LOW See supportlist	50	VOFSUPP-PRIM-PSK-ATTA	1

PIPE SPOOLS

[2]



Connection to existing pipe to be checked on site

For installation tolerances dimensions to be checked by pipe contractor at site.
Degree of prefabrication to be decided by contractor.

NOTE:

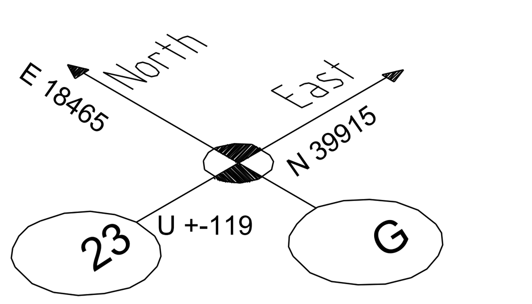
Site weldings to be added by contractor.

Buildings construction and equipment erection tolerances to be checked by contractor.

Pipe levels and dimensions are according to pipe center line.

Isometric drawings for sizes DN 15-25 are approximate.
Piping contractor is responsible for final routing and supporting.

ORIGO & MARKINGS



1 DENOTES PARTS LIST
1 DENOTES WELD NO.
1 DENOTES REVISION
INSULATION
INSUL. + TRACE.
BRANCH START POINT

Insulation table (lengths according SSG 7645 Alt2)

Bore	Thickness	Length	Insulation spec name
50	20mm	5897mm	/PSK3704-6a-Prot-alum
40	20mm	153mm	/PSK3704-6a-Prot-alum
80	20mm	2911mm	/PSK3704-6a-Prot-alum
15	20mm	182mm	/PSK3704-6a-Prot-alum
150	60mm	286mm	/PSK3704-6a-Prot-alum

Welding factor	Acc. to pipe class	Operating temp.	103degC
Effiency of long. weld	Acc. to pipe class	PED Module	-
NDT circ. weld %	-	PED Group	-
NDT long. weld %	-	Wind loads	-
Welding list see doc.	-	External loads	-

		Hazardous CLP content	
Directive	2014/68/EU	Pipe Class	/119440-VE16H1A-R00
Design press. (min)	-1bar	PED Category	SEP
Design temp. (min)	-	Media	
Max. allow. press. (PS)	-	Standard	
Operating press.	0.1bar	P&ID	RAU8F00303
		Design press.	6bar
		Design temp.	220degC
		Test press.	-
		Pipe list doc. ID	RAUAF04300
		Valve list doc. ID	RAUAF04301

Rev	Revision Note	Created by	Checked by	Approved by	Date
0		J. Lakka	T. Saarinen	M. Kankkunen	5/5/2026

Site: Tervasaari, Valkeakoski
Equipment Type: Höyry- ja lauhdejärjestelmä
Equipment ID:
Project Key: TERPM5_N5265A
Project Name: Ter PM 5 Safe Future
Project Number: N5265A
Work:
Projection Method
Welding Symbols: ISO 2553
General Tolerances: ISO 13 920-BF ISO 2768-mK-E ISO 2768-mK-E ISO 2768-mK-E ISO 2768-mK-E
Scale: -
Size: A1
Weight: -

Valmet

Title: 50216-VLA-80-16H1A

Status: CERTIFIED
Document ID:
Security Classification: Internal

Revision 0